

Skills

Product: Design Thinking, Agile Methodology, Sprint Planning, Product Testing & Lifecycle Management, User Research

Development: JavaScript, React, C#, Python, C++, Java, HTML, CSS, Unity, Arduino, SASS

Tools/Technologies: GitHub, Linear, Microsoft Azure, Figma, SolidWorks, MATLAB, LTSpice, FEA, BioPac

Experience

QA Automation (Product Management Contributor) | Scispot.io

Sept 2024 - Dec 2024

- Led **feature prioritization**, **bug detection** and **ticket management** for 700+ issues across development, DevOps, and design teams, ensuring high-impact bugs and features were addressed on time within **1-week Agile sprints** using **Linear**.
- Authored & maintained **130+ test cases** (manual & automated). Integrated **CI/CD workflows** into development process.
- Contributed to **sprint planning** and **product roadmap** in a start-up environment. Worked with development, customer support, design to ensure releases meet technical and client expectations.
- Collaborated with **17+ clients** in calls and communications, sharing technical progress and gathering actionable feedback that directly influenced the product roadmap, **improving customer satisfaction by 25%**.

Software Research Assistant | PhotoMedicine Labs

Jan 2024 - April 2024

- Spearheaded development and design of responsive website using **JavaScript**, **React**, and **HTML/CSS** to enable users to view and analyze high-resolution tissue sample images, improving research teams' ability to make data-driven decisions.
- Integrated **OpenSeadragon API** and cloud services such as **Microsoft Azure Blob Storage** and **Cosmos DB**, enabling smooth data access and improving the efficiency of data storage and retrieval.
- Piloted the design and implementation of key product features, including **fullscreen**, **split screen**, and **synchronized viewers**, enhancing user interaction and ensuring easy comparison of complex data.

Student Investigator | Watolink Design Team, Symphonic Labs

Sept 2023 - Dec 2023

- Led the design and execution of a research study with a team of 4, taking ownership of **user research**, **data collection strategies**, and focusing on **human factors** to inform design decisions of a wearable prosthetic for speech translation.
- Explored appropriate texts, electrode placement, and facial muscles required for speech to collect data for a wearable prosthetic that translates facial muscle contractions for speech.
- Aided in screening of **70+ human participants** to ensure a diversity; performed **20+ procedures** on internal members.

Projects

MindFire | BCI-Controlled Game Development

- Collaborated in a team of 8 to design game storyline and visuals, taking input from keyboard controls and user's blinks via a brain-computer interface (BCI), **Neurocity Crown**.
- Designed and developed key game features, including **user interface design** (start menu) and core gameplay functionalities in **Unity** and **C#**. Also named the game!
- Achieved **Top Ten Finalist** at the **2024 IEEE GameSIG Showcase**, demonstrating successful product development, effective team collaboration, and ability to deliver innovative solutions under tight deadlines.

EMG-Controlled Water Gun Prosthetic | SolidWorks, Circuit Design, Software Integration

- Designed and modeled a functional prototype of a water gun prosthetic hand, integrating mechanical components, **Arduino**-based control in **C/C++**, and **EMG sensors** to enable precise, real-time movement control.
- Conducted mechanical calculations for structural strength, circuit simulations in **LTSpice**, and performed **motion analysis simulations** in **SolidWorks** to make data-driven improvements to actuators and overall mechanical performance.
- Defined and **prioritized key features**, such as actuator performance and user functionality. Iterated the prototype based on user testing and engineering feedback, refining the design to improve **usability** and **user experience**.

Education

University of Waterloo | Bachelor of Applied Science, Honours Biomedical Engineering

Sept 2022 - April 2027

- **Relevant Courses:** Intro to Biomedical Design, Human Factors in Biomedical Design, Digital Computation (C++), Data Structures & Algorithms (C++), Matrices & Linear Systems (MATLAB), Visual Communications (SolidWorks)
- President's Scholarship of Distinction